

FAILSAFE

AI-Powered Student Failure Risk Prediction System

Abstract

FAILSAFE is an end-to-end AI-powered web application that predicts student academic failure risk early in the semester using machine learning and Explainable AI. Built on XGBoost and SHAP, it flags at-risk students before final examinations, explains every prediction in plain English, and auto-generates personalised intervention plans — enabling faculty to act before it is too late.

PROBLEM STATEMENT Traditional academic monitoring detects failure only after final grades are published, leaving no time for corrective action. Faculty lack proactive tools to identify struggling students mid-semester, understand root causes, or deliver targeted support. FAILSAFE solves three specific problems: - Predict failure risk early using grades, absences, and behavioural data. - Explain every prediction using SHAP so faculty understand and trust the results. - Act by auto-generating personalised intervention plans per student.	DEVELOPMENT APPROACH The system is built across four integrated layers: - ML Pipeline — data loading, XGBoost training, SHAP computation. - Backend — FastAPI with JWT auth, SQLAlchemy ORM, PostgreSQL, role-based access, python. - Database — PostgreSQL with soft-delete (is_deleted) across all tables. - Frontend — React.js with Pie charts, SHAP bar charts, intervention tracking. DATASET UCI Student Performance Dataset — student-mat.csv
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Model Details & Performance Metrics

~90%	~0.94	316	79	33
Accuracy	ROC-AUC	Train Samples	Test Samples	Input Features

RISK SCORE INTERPRETATION

Risk Level	Score Range	Meaning	Action Required
HIGH RISK	>= 65%	Student very likely to fail final exam	Immediate counselling + interventions
MEDIUM RISK	35 - 65%	Elevated risk, needs monitoring	Support measures + weekly tracking
LOW RISK	< 35%	Performing within acceptable range	Continue monitoring each semester

SHAP Explainability Insights & Key Features

TOP SHAP RISK FACTORS IDENTIFIED

+0.310 FAILURES Past academic failures are the single strongest predictor. Each additional failure substantially raises predicted risk.	+0.120 STUDY TIME Students studying less than 2 hrs/week show elevated risk. Insufficient preparation correlates with poor outcomes.
+0.240 ABSENCES Students missing more than 15 classes show markedly elevated risk. Absenteeism directly impacts content retention.	+0.095 HIGHER EDU Lack of aspiration for higher education reduces motivation. Students answering no show measurably higher risk.
+0.210 G2 GRADE Second period grade is a leading indicator. Scores below 8/20 in G2 strongly predict final failure.	-0.065 FAMILY REL Strong family relationships provide emotional and academic support, acting as a consistent protective factor.
+0.180 G1 GRADE First period grade provides the earliest measurable signal of academic trajectory within the semester.	-0.045 INTERNET Home internet access enables self-directed study and resource access, marginally reducing failure probability.

SHAP TO INTERVENTION MAPPING

SHAP Factor Detected	Auto-Generated Intervention	Type
High failures score	Structured remedial study plan with weekly milestones	Study Plan
High absences score	Daily attendance monitoring + family notification protocol	Attendance
Low G1 or G2 grade	Peer mentor assignment + practice exam papers with solutions	Academic Support
High alcohol consumption	Wellness check + lifestyle counselling referral	Wellness
No higher edu aspiration	Career counselling + alumni mentorship connection	Motivation
Risk score >= 65%	Immediate one-on-one counselling session with faculty	Counselling

KEY FEATURES IMPLEMENTED

Early Risk Prediction XGBoost predicts failure probability from G1, G2, absences, and behavioural data mid-semester.	SHAP Explainability Per-student SHAP values stored in DB and visualised as colour-coded bar charts in the UI.
AI Intervention Plans Rule engine maps SHAP factors to counselling, study plans, attendance monitoring, and wellness.	Bulk CSV Upload Faculty upload UCI-format CSV files to import and auto-predict entire cohorts at once.
JWT Authentication Secure role-based access with 24-hour tokens. Faculty see own students; HOD sees all.	Soft Delete is_deleted column on all tables ensures data is archived not destroyed, maintaining audit trails.
Dark / Light Mode Theme preference stored in PostgreSQL settings table and restored automatically on every login.	Edit & Re-predict After editing student data the model automatically re-runs to reflect updated information.

Conclusion: FAILSAFE demonstrates how machine learning and explainable AI can transform academic monitoring from a reactive end-of-semester activity into a proactive, data-driven support system. By combining XGBoost's predictive accuracy with SHAP's transparency and an automated intervention engine, the system gives faculty not just a warning — but a complete understanding of each at-risk student's situation and a concrete, personalised plan of action to act before it is too late.

Disclosure : Report has been generated by Generative AI (Claude) and edited by me (Naitik Gupta) as per the project requirements.